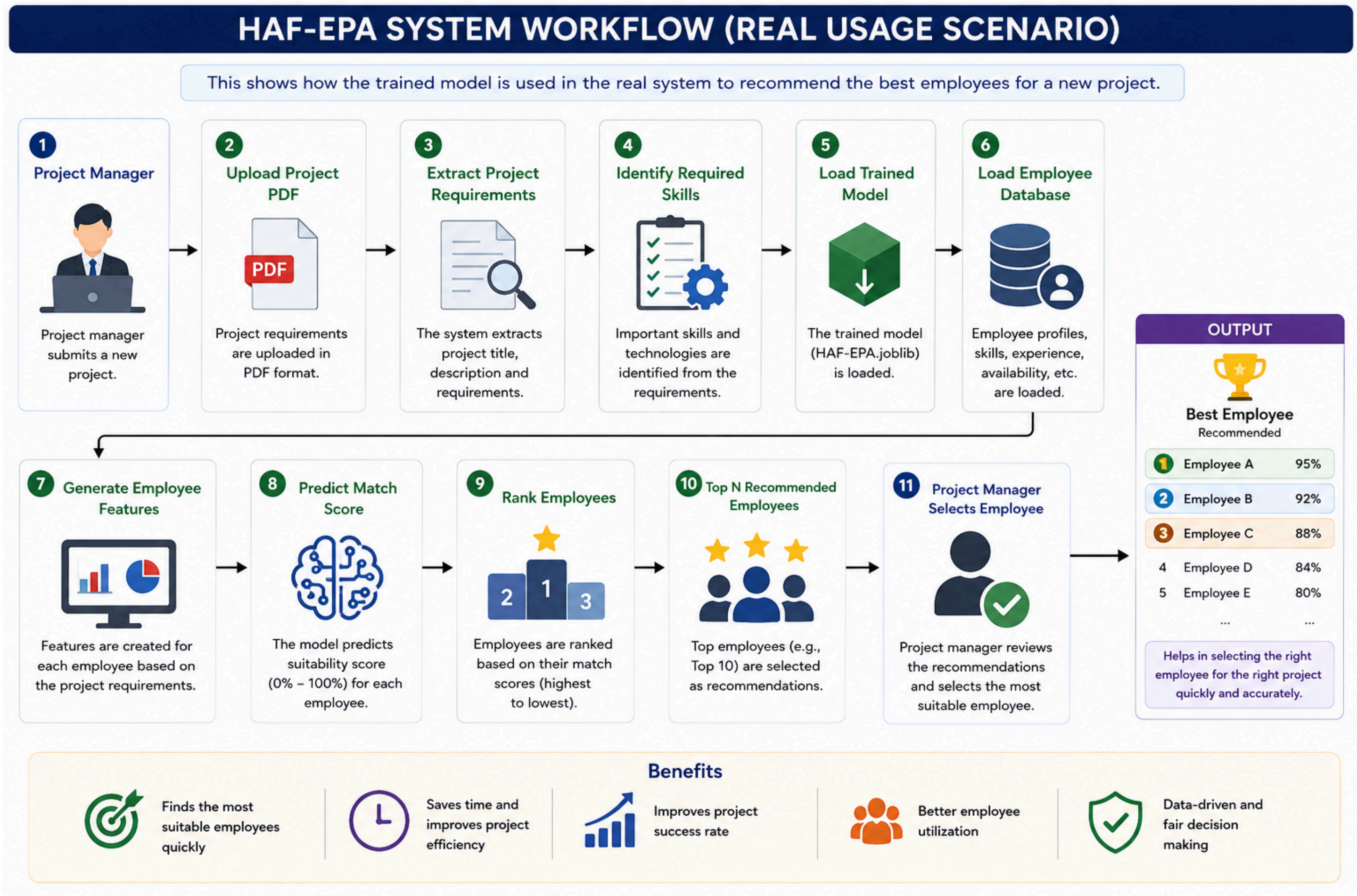

HAF-EPA Web Application

After successfully training and validating the HAF-EPA machine learning model, the model is integrated into a Flask-based web application. The purpose of this web application is to help project managers quickly identify the most suitable employees for a new software project.

Step 1: Project Manager Uploads a Project PDF

The process begins when a project manager uploads a project requirement document in PDF format through the web application.
The uploaded PDF typically contains:

- Project Overview
- Required Skills
- Responsibilities
- Technologies
- Experience Requirements
- Project Objectives

Step 2: PDF Validation and Text Extraction

Once the PDF is uploaded, the system validates whether the document contains a valid software project description.

After validation, the application extracts all textual information from the PDF, including project requirements, technologies, responsibilities, and skill-related information. This converts the unstructured project document into machine-readable text.

Step 3: Requirement and Skill Identification

The extracted text is analyzed to identify the project's required competencies. The system categorizes the extracted requirements into two major groups:

Programming Skills

Technical skills required for software development, such as:

- Python
- Java
- JavaScript
- React
- Flask
- SQL
- Docker
- Machine Learning

Management and Soft Skills

Skills related to project coordination and teamwork, such as:

- Project Management
- Agile Methodology
- Scrum
- Leadership
- Communication
- Team Collaboration

Since this recommendation system is designed for software companies, both technical expertise and management capabilities are considered important factors.

Step 4: Data Normalization and Preparation

The extracted skills and project requirements are standardized and normalized to ensure consistency with the data used during model training.
During this stage:

- Duplicate skills are removed.
- Skill names are standardized.
- Project requirements are structured into a format compatible with the trained model.
- Programming and management skill groups are prepared for comparison against employee profiles.

This ensures that the prediction process follows the same methodology used during training.

Step 5: Load the Trained Model
The web application loads the trained machine learning model:

HAF-EPA.joblib
This model was previously trained using historical employee-project assignment data and has already learned the patterns associated with successful employee recommendations.

Step 6: Load Employee Reference Database
The system also loads the employee reference dataset:

HAF-EPA_employee_reference.csv
This dataset contains information about employees, including:

- Employee Name
- Skills
- Experience
- Availability
- Historical Performance
- Programming Skill Information
- Management Skill Information

Step 7: Generate Employee–Project Features
For every employee in the reference database, the system compares the employee profile with the uploaded project's requirements.

Several predictive features are generated, including:

- Skill Match Score
- Programming Skill Match
- Management Skill Match
- Experience Score
- Availability Score
- Performance Score
- Compatibility Score
- Project Requirement Coverage

Step 8: Apply the HAF-EPA Machine Learning Model
The generated feature set is passed to the trained HAF-EPA model.
The model evaluates every employee individually and predicts a suitability score representing the likelihood that the employee is a good fit for the project.
The higher the score, the better the employee matches the project requirements.

Step 9: Rank and Shortlist Employees
After predictions are generated, employees are ranked from highest to lowest suitability score.
The system automatically creates a shortlist of the top recommended employees.

Typically, the top 10 employees are selected as final recommendations.

Step 10: Display Results and Visualizations
Finally, the web application presents the recommendation results to the project manager.
The results include:

- Recommended Employee List
- Match Percentage
- Programming Skill Match
- Management Skill Match
- Experience Information
- Availability Information
- Interactive Graphs and Charts

These visualizations help project managers make informed and data-driven staffing decisions.